Name: _____ Class: _____ No.: _____

Check your measurements and analysis

Distribute only after B is completed and first answers preserved. Check, then correct in another colour without erasing the original.

Plate	Sample	Visible zone outside the disc?	Total clear-zone（清晰區） diameter (mm)
1	X	Yes	18.7
	Y	No	0.0
	Z	Yes	26.1
	C (control)	No	0.0
2	X	Yes	18.7
	Y	No	0.0
	Z	Yes	26.3
	C (control)	No	0.0
3	X	Yes	18.7
	Y	No	0.0
	Z	Yes	26.5
	C (control)	No	0.0

Mean total diameter calculation	X	Y	Z	C (control)
mm	18.7	0.0	26.3	0.0

Reference values are not a reason to overwrite valid readings. Ruler readings may differ within ±1 mm; check means against your own readings.

1. What does the appearance of a clear zone（清晰區） indicate?	B. Under these test conditions, there is no visible bacterial growth in the zone, indicating inhibition（抑制）.
2. What does the diameter of a clear zone（清晰區） represent?	C. The extent of the area with no visible bacterial growth under these conditions; diffusion and other factors also affect the diameter.
4. Why do experiments need to be repeated?	C. To reduce random error and assess the reproducibility（可重複性） of the results and the reliability（可靠性） of the data.
5. If repeated results differ greatly, what should you do, and why?	D. First check spreading, disc positions and plate conditions, then repeat as needed to identify causes of variation and improve the comparison.
6. When comparing the clear zones（清晰區） around C (control) and X, Y and Z, what is the main role of C?	C. Provide an antibiotic（抗生素）-free comparison to help assess whether clear-zone（清晰區） formation is associated with the antibiotic（抗生素） sample.

3. Ranking of mean total clear-zone（清晰區） diameter under these conditions: Z ＞ X ＞ Not applicable ＞ Not applicable。

Samples with external zones: X、Z。

Check reasoning and correct

Distribute only after B is completed and first answers preserved. Check, then correct in another colour without erasing the original.

Your initial observations

A clear zone（清晰區） with no visible bacterial growth is present near Penicillium（青黴菌）(a fungus). This is an observation; its cause needs testing.

Independent variable	Sample in the disc
Dependent variable	Visible clear zone（清晰區） and its total diameter
Controlled variables	Disc size ／ Composition of the agar plate（瓊脂板） ／ Type of bacteria used ／ Initial bacterial distribution ／ Preset sample preparation
What are the assumptions of this investigation?	Bacteria are initially distributed evenly on each agar plate（瓊脂板）. ／ Equipment does not introduce contamination. ／ All plates use the same simulated observation conditions.

Use a matching carrier-only disc without antibiotic（抗生素） to check carrier/disc effects, with other conditions the same. Call it a water control only if every carrier is water.

Hypothesis evaluation/application

B is testable because it compares a sample/carrier under shared conditions using zone presence and diameter. Accept a reasoned conditional hypothesis for U and another bacterium, without requiring an outcome guess.

Labels, spacing and errors

Use X/Y/Z/C once each, in any order. Space discs apart and away from the edge to avoid overlap and difficult interpretation. A risks contamination; B changes bacteria and makes the comparison unfair; C is appropriate.

Repeat-plan comparison

Plan B uses new independently prepared plates to compare variation. Remeasuring one plate estimates reading variation, not independent experimental variation. Three is a classroom choice; balance resources, time and useful evidence.

Procedure and errors

Order: R → U → Q → T → S → P. A: check even spreading. B: prepare a new independent plate rather than call a second reading a new result. Retain differing data and investigate causes.

Concept application

For example, X has readings 18.7, 18.7, 18.7 mm and an external zone. Pre-existing resistant（具抗藥性的） bacteria are more likely to survive and reproduce（繁殖）; traits can be inherited（遺傳）. Their proportion may rise, so effectiveness may fall without a complete takeover. Accept Z with its own readings. The experiment does not prove resistance（抗藥性） arose.

My correction and reason (if originally correct, state the check used)

Correct the evidence or causal link in your concept application.